Unplugged Activity
A pile of Pennies

Unplugged Activity : Description

Goal: find oldest penny.

We are going to explore several solutions to achieving the goal.

Unplugged activity description

I have 50 coins, and the problem is to find the oldest one in the quickest time. So, we are going to do it first in a sequential mode and then in various parallel variations.

Students break into groups of 5 or 6 or as instructed by the instructor.

Phase 1 (sequential): each group (student 1 sorting and student 2 timing, other students observing)
Student 1 task: find the oldest penny
Student 2 task: time the activity of student 1 and writes it down.
Other students' task: observe Student 1 and 2. When they are done **Shuffle coins and put back in bag**

Unplugged activity description cont'd….

Phase 2 (Parallelize it): in each group (students 1 and 2 are given equal number (25) of pennies and are asked to find the oldest penny in their pile, student 3 records the times for student 1 & 2. Parallel Time is calculated as the time of the slowest student among 1 and 2.

At a set time, sorting begins for all groups.

Student 3 records the time it takes each student to find the oldest penny in their pile. Parallel Time is calculated as the time of the slowest student among 1 and 2.

When done **shuffle coins and put back in their respective bags. Please do not mix coins. Each bag should have 25 pennies.**

Unplugged activity description cont'd….

Phase 3 (Parallelize it-2):

Parallelize it (2): Teacher gives 2 pennies to students 1, 2, 3, 4 and the remaining pennies to student 5. If less than 5 students per group, then one student gets remaining pennies after 2 each have been distributed to other students in the group. Have one student spare for timing purposes.

At a set time, sorting begins for all groups.

Student 6 records time for each of the worker students. Parallel time is calculated as the time of the slowest student.

What do we learn? **This is poor load balancing and leads to a higher overall parallel execution time!**

Shuffle and put coins back in the bag

Group Discussion:

What did we learn?

Survey link

https://forms.gle/SYBfF4ihFvNiTk7q7

Thank you for your participation!

Hope you enjoyed the parallelism day